SHIPMENT & DELIVERY PERFORMANCE KPI ANALYSIS

Project Type	Logistics / Supply Chain KPI Analysis
Primary Tool	Microsoft Excel
Analysis Period	09-Jan-2025 to 15-Jan-2026
Records Analyzed	2,429 shipments
Core Focus	On-Time, In-Full, OTIF & Carrier Performance

1. Business Context & Analytical Objective

In logistics, a shipment is successful only when the customer receives the required quantity within the expected timeframe. Looking at shipment volume alone can hide service problems, while looking at one KPI can make it difficult to identify the real operational priority.

This project was therefore designed to answer a practical management question: **Where is shipment performance breaking down, and what should the business investigate first?**

Objectives

- Measure total shipment activity and the main delivery KPIs.
- Evaluate On-Time Delivery and identify the scale of late shipments.
- Measure In-Full performance and understand quantity-fulfillment quality.
- Calculate and interpret OTIF performance.
- Break OTIF failures into four operationally meaningful categories.
- Compare carrier performance using OTIF and average delay.
- Translate analytical findings into practical business recommendations.

2. Dataset & Scope

The analysis covers shipment-level records from **09-Jan-2025 through 15-Jan-2026**. The working dataset contains 2,429 shipments and includes dates, quantities, carrier information, and indicators used to evaluate delivery performance.

Field	Analytical purpose
ShipmentID	Unique shipment tracking
OrderID	Order-level reference
SupplierID	Supplier reference
ShipDate	Shipment timing
PromisedDate	Expected delivery date
ActualDeliveryDate	Actual delivery timing
Carrier	Carrier comparison
QtyShipped / QuantityOrdered	In-Full assessment
Lead_Time	Shipment cycle measurement
Delay	Delivery delay measurement
OT / IF / ONIF	On-Time, In-Full and OTIF indicators

3. KPI Framework & Results

The KPI layer was built to move from a high-level performance view to the specific driver of service failure. The final dashboard presents six core KPIs so a reviewer can understand the situation quickly.

KPI	Result	Business interpretation
Total Shipments	2,429	Shipment volume in the analysis period
On-Time Delivery	23%	Only a minority met the promised delivery timing
In-Full	82%	Quantity fulfillment is comparatively stronger
OTIF	20%	Only about one-fifth met both requirements
Average Lead Time	10.42 days	Overall shipment lead-time level
Average Delay	6.92 days	Significant delivery-delay exposure

How the numbers connect

The most important analytical observation is the gap between **82% In-Full** and **23% On-Time**. This means quantity fulfillment is not the dominant weakness in the available data. The much lower On-Time result points toward delivery timeliness as the major service problem.

The average lead time of 10.42 days and average delay of 6.92 days reinforce the need to examine the delivery process, not just the final OTIF percentage.

3.1 Why OTIF Matters

OTIF combines the two customer-facing requirements: a shipment must be **On-Time** and **In-Full**. A shipment that satisfies only one condition still fails the combined service measure. This makes OTIF useful for management because it captures the complete delivery promise.

3.2 OTIF Failure Breakdown

Category	Shipments	Meaning
OTIF	475	On-time and in-full
Late + In-Full	1,506	Full quantity, but late
On-Time + Not In-Full	90	On time, but incomplete
Late + Not In-Full	358	Both requirements failed

Analytical conclusion: Late + In-Full is by far the largest category. The business therefore has a clearer opportunity in improving delivery timeliness than in treating all OTIF failures as quantity problems.

4. Carrier Performance Analysis

Once the overall problem was identified, the next question was whether carrier performance differed meaningfully. Five carriers were compared using OTIF percentage and average delay.

Carrier	Shipments	On-Time %	OTIF %	Avg Delay (days)
DHL	511	24%	21%	5.14
FedEx	446	24%	20%	4.50
Ekart	475	23%	20%	4.65
BlueDart	496	25%	20%	4.76
Delhivery	501	20%	18%	5.19

4.1 What stands out?

Delhivery has the weakest OTIF result at 18% and the highest average delay at 5.19 days. It also has the lowest On-Time percentage at 20% in the carrier table. This combination makes Delhivery the clearest carrier-level signal for further investigation.

DHL has the highest OTIF at 21% among the five carriers. FedEx has the lowest average delay at 4.50 days. The differences are not enormous, so the analysis should be used to prioritize questions rather than to make an immediate carrier replacement decision.

4.2 Analytical Caution

Carrier performance can be affected by shipment mix, routes, destinations, service levels, volumes, and other operational factors. The current analysis does not control for all of those variables. Therefore, the correct business interpretation is: **Delhivery should be reviewed first**, not **Delhivery is definitively responsible for the overall delay problem**.

4.3 Why the Carrier Chart Was Changed

The dashboard originally considered carrier OTIF performance, but the final carrier visual focuses on **Average Delay by Carrier (Days)**. This provides a different analytical angle from the overall OTIF chart and directly highlights where delivery delays are highest. The final visual uses a simple blue bar treatment with white data labels for readability.

4.4 Carrier Ranking

A ranking was also calculated for carrier OTIF performance. The ranking is useful as a supporting analysis, while the dashboard prioritizes the more decision-oriented delay comparison.

Carrier	OTIF %	Rank
DHL	20%	2
FedEx	20%	3
Ekart	20%	4
BlueDart	21%	1
Delhivery	18%	5

5. Dashboard Development & Business Story

The dashboard was designed around the way a manager or recruiter would scan a portfolio project: first understand the scale, then identify the problem, then see where attention should go.

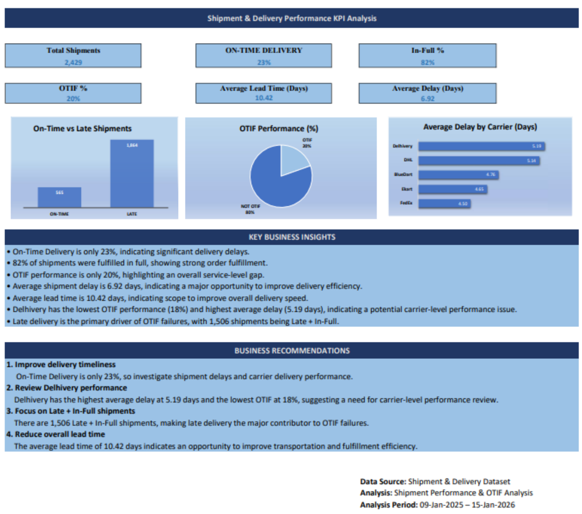
Dashboard element	Purpose
Total Shipments	Establishes analysis volume
On-Time Delivery	Shows the main timeliness issue
In-Full	Shows quantity-fulfillment strength
OTIF	Summarizes combined service performance
Average Lead Time	Shows shipment-cycle level
Average Delay	Quantifies delay exposure
On-Time vs Late Shipments	Makes timeliness gap visible
OTIF Performance (%)	Communicates overall service result
Average Delay by Carrier	Identifies carrier-level delay differences
Key Business Insights	Explains what the charts mean
Business Recommendations	Connects findings to action

5.1 Design Decisions

- The dashboard uses a limited number of relevant charts rather than filling the page with visuals.
- KPI cards are placed at the top so the main results can be understood immediately.
- Carrier delay is presented separately from overall OTIF to avoid repeating the same message.
- Blue bars and white data labels improve consistency and readability.
- Business Insights and Business Recommendations are included so the workbook demonstrates interpretation, not only chart creation.
- The dashboard was formatted for a one-page landscape print layout for portfolio sharing.

5.2 From Data to Decision

Raw shipment records → KPI calculations → OTIF breakdown → carrier comparison → insights → recommendations.



6. Key Insights, Recommendations & Conclusion

6.1 Key Business Insights

- **Delivery timeliness is the main weakness:** On-Time Delivery is only 23%.
- **Quantity fulfillment is comparatively strong:** In-Full performance is 82%.
- **OTIF is weak:** overall performance is approximately 20%, showing that customers frequently miss at least one service requirement.
- **Late + In-Full dominates:** 1,506 shipments fall into this category, making late delivery the most important failure pattern.
- **Delhivery needs review:** it has the lowest OTIF (18%) and highest average delay (5.19 days) among the analyzed carriers.
- **Lead time is an improvement opportunity:** average lead time is 10.42 days and should be monitored alongside delay.

6.2 Business Recommendations

1. Improve delivery timeliness — Investigate the causes behind late shipments, including dispatch timing, transit delays, handoffs, routing, and other available operational factors.

2. Review Delhivery performance — Use the carrier result as a trigger for a deeper review of service levels, routes, delay reasons, and shipment mix before making allocation decisions.

3. Prioritize Late + In-Full shipments — Because this is the largest OTIF failure group, reducing this population should be a high-priority improvement area.

4. Reduce overall lead time — Review fulfillment and transportation stages to identify avoidable waiting time and process bottlenecks.

5. Create recurring carrier scorecards — Track OTIF, On-Time %, average delay, and shipment volume over time so management can monitor changes rather than relying on a one-time snapshot.

6.3 Limitations & Recommended Next Analysis

The current project is a descriptive analysis. It identifies patterns but cannot establish causation. For a stronger operational study, additional fields such as delay reason, origin, destination, warehouse, route, product category, supplier, transportation mode, and service type could be analyzed if available.

A future version could also add monthly or weekly trends, carrier comparisons adjusted for shipment mix, and root-cause analysis of the highest-delay segments.

6.4 Conclusion

The completed analysis tells a consistent business story: the organization is comparatively better at fulfilling quantities than at meeting promised delivery dates. The 82% In-Full result contrasts with the 23% On-Time result, producing an OTIF performance of approximately 20%.

The OTIF breakdown makes the priority clear: 1,506 shipments were Late + In-Full. Carrier analysis then provides a practical place to start, with Delhivery showing both the lowest OTIF and the highest average delay. The recommended response is therefore to improve delivery timeliness, investigate the largest failure group, and establish consistent carrier performance monitoring.

Data Source: Shipment & Delivery Dataset | Analysis: Shipment Performance & OTIF Analysis | Period: 09-Jan-2025 – 15-Jan-2026